# BIJOY SARKAR

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| Date of Birth | Nationality | Gender | LinkedIn | GitHub |
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## PROFILE

Computer Science & Engineering undergraduate with hands-on experience in AI/ML, full-stack web development, cybersecurity, and mobile application development. Experienced in Python, Flutter, Firebase, computer vision, NLP, and software development through internships and practical projects. Currently seeking software development and AI/ML opportunities to build production-oriented solutions and strengthen problem-solving skills.

## EDUCATION

**Bachelor of Technology in Computer Science and Engineering** **Aug 2023 – Present**
*ICFAI University Tripura, Agartala*

## WORK EXPERIENCE

**Cybersecurity Intern** **Jun 2026 – Jul 2026**
*ETDC, STQC Directorate, MeitY, Government of India*

- Worked on cybersecurity frameworks and Generative AI security with focus on security assessment of AI applications.
- Conducted security risk analysis using **OWASP guidelines for Large Language Models (LLMs)** to identify potential vulnerabilities and mitigation strategies.
- Gained practical exposure to vulnerability assessment, AI application security, and IT testing and quality assurance practices.

**AI/ML Intern** **Jan 2026 – Mar 2026**
*National Institute of Technology Silchar*

- Worked on a **Road Segmentation** project using satellite imagery for identifying and extracting road networks.
- Performed image preprocessing, dataset preparation, feature extraction, and semantic segmentation using Python.
- Worked with satellite imagery datasets and evaluated computer vision models for road detection and segmentation.

**App Developer Intern** **Jun 2025 – Jul 2025**
*TalkEng, Agartala*

- Developed an **Expense Tracker** mobile application using **Flutter and Dart**.
- Implemented Firebase Authentication and real-time database integration for secure user data management.
- Worked on application functionality, user interface development, and overall user experience.

**Full-Stack Development Intern** **Jun 2024 – Aug 2024**
*Apni Vidya, Agartala*

- Contributed to both front-end and back-end development of web applications.
- Worked on application design, coding, testing, debugging, and deployment-related activities.
- Collaborated on feature development and improved application functionality through practical development tasks.

**Computer Tutor** **Mar 2022 – Aug 2022**
*Novel Computer, Agartala*

- Taught computer fundamentals, programming basics, MS Office, and practical IT skills through hands-on learning.

## TECHNICAL SKILLS

**Languages:** Python, C++, C, Java, JavaScript, Dart, SQL

**AI/ML:** Machine Learning, Deep Learning, Computer Vision, NLP, OpenCV, Jupyter Notebook

**Web & App:** HTML, CSS, Flask, Streamlit, Flutter, Firebase

**Databases:** MySQL, SQLite, Firestore

**Cybersecurity:** OWASP, LLM Security, Vulnerability Assessment, AI Security

**Tools:** Git, GitHub, Android Studio, VS Code

## PROJECTS

**OWASP AI Security Assistant** **[GitHub]**
*Python Flask | Ollama | Llama 3 | Docker | OWASP*

- Developed a cybersecurity web application to assess code, URL endpoints, and infrastructure against OWASP vulnerability categories.
- Integrated local AI inference using **Ollama and Llama 3** for security analysis, vulnerability assessment, and remediation recommendations.
- Implemented assessment history, security metrics, dataset management, re-evaluation workflows, and Docker Compose deployment.

**Rakshak-AI** **[GitHub]**
*Python | LM Studio | Llama 3.2 | Tkinter | SQLite*

- Developed a locally hosted AI assistant using **Python** and LM Studio.
- Integrated conversational assistance, desktop GUI, BMI calculation, speech recognition, and text-to-speech functionality.
- Used **SQLite** for persistent data management and an interactive assistant workflow.

**Road Segmentation from Satellite Images** **[GitHub]**
*Python | Computer Vision | Jupyter Notebook | Kaggle*

- Developed a computer-vision workflow for identifying and extracting road networks from satellite imagery.
- Performed image preprocessing, dataset preparation, model training, and evaluation.
- Explored semantic segmentation for geographic mapping and infrastructure planning.

**Expense Tracker** **[GitHub]**
*Flutter | Dart | Firebase | Firestore*

- Developed a mobile application for recording and categorizing daily expenses.
- Integrated **Firebase Authentication and Firestore** for secure user authentication and real-time data management.
- Implemented expense summaries and an intuitive interface for monitoring personal spending.

**Smart Attendance System Using Face Recognition** **[GitHub]**
*Python | OpenCV | Face Recognition | Computer Vision*

- Developed an automated attendance system using **face recognition and computer vision**.
- Implemented face detection, feature extraction, identity matching, and automated attendance recording.
- Designed organized attendance history with secure administrative access and device-token checks.

**CivicLens Core** **[GitHub]**
*FastAPI | PostgreSQL | PostGIS | Redis | Docker | Next.js*

- Built a full-stack public-infrastructure intelligence platform for mapping assets and reported civic issues.
- Designed API, database, geospatial, caching, background-worker, and object-storage components with Docker.
- Structured Part 1 to support future AI-based risk prediction, prioritization, hotspot detection, and maintenance recommendations.

**MiniC Compiler Studio** **[GitHub]**
*C/C++ | Compiler Design | Python | GCC | Docker*

- Built a web-based compiler environment for a MiniC language with a browser-based development interface.
- Integrated lexical analysis, parsing, compilation workflows, GCC execution, and source-code processing.
- Containerized the application and exposed the development studio through a Dockerized web server.

**PixelDB Engine** **[GitHub]**
*Java | Maven | Database Systems | SQL*

- Developed a database engine from scratch to explore storage, query-processing, and database-system internals.
- Used Java and Maven with automated tests and a structured engine architecture.
- Planned a pixel-art themed interface to make database concepts more interactive and demonstrable.

**Advanced NLP Analytics** **[GitHub]**
*Python | Flask | NLP | TF-IDF | Logistic Regression | Docker*

- Built an end-to-end NLP application for text classification using preprocessing, tokenization, unigram/bigram TF-IDF features, and Logistic Regression.
- Added a Flask interface for real-time text analysis and CSV profiling with confidence scores and keyword extraction.
- Added automated tests, Docker support, and GitHub Actions-oriented project structure for reproducible development.

**AI-Based Text Sentiment Analysis — SentimentX** **[GitHub]**
*Python | Flask | NLTK | VADER | YouTube API*

- Developed a web-based sentiment analysis system using **VADER** and NLTK for positive, negative, and neutral classification.
- Integrated YouTube comment analysis through the YouTube API and automated scanning of comment samples.
- Built the application with Flask and structured it for containerization and GitHub-based deployment workflows.

**Multithreaded Real-Time LAN Chat** **[GitHub]**
*Java | TCP Sockets | Multithreading | Maven*

- Built a real-time LAN chat application using Java networking and socket-based client-server communication.
- Used multithreading to handle multiple clients concurrently while maintaining responsive message delivery.
- Structured the project with separate server and client components and Maven-based build configuration.

**NovaMart — Multi-Vendor Commerce Platform** **[GitHub]**
*Next.js | TypeScript | Prisma | SQLite | Razorpay*

- Designed a full-stack multi-vendor commerce platform with product, checkout, seller, and customer workflows.
- Used Next.js, TypeScript, Prisma, and SQLite for application and database development.
- Integrated payment-oriented architecture and handled frontend animation and checkout implementation challenges.

**Facebook Sentiment Analysis** **[GitHub]**
*Python | NLP | Pandas | Machine Learning*

- Explored sentiment classification of social-media text using Python-based NLP and machine-learning workflows.
- Performed text preprocessing, feature preparation, model experimentation, and sentiment interpretation.
- Used data-analysis workflows to study patterns in user-generated textual content.

**Fruit Image Recognition** **[GitHub]**
*Python | Computer Vision | Deep Learning | Image Classification*

- Developed an image-recognition workflow for classifying fruit images using computer-vision techniques.
- Worked on image preprocessing, dataset preparation, model training, and evaluation during AI/ML project work.
- Applied deep-learning concepts to visual classification and model-performance analysis.

**AI/ML and NLP Experiment Portfolio** **[GitHub]**
*Python | Jupyter | OpenCV | NLTK | Matplotlib | Kaggle*

- Built and experimented with practical AI/ML workflows covering computer vision, NLP, sentiment analysis, and dataset-driven modelling.
- Used Jupyter Notebook and Kaggle datasets for preprocessing, visualization, experimentation, and model evaluation.
- Maintained reusable project structures and documented workflows for reproducible development and learning.

## ADDITIONAL PROJECTS

**AI/ML Internship — Road Segmentation** **[GitHub]**
*Python | Computer Vision | Satellite Imagery | Kaggle*

- Applied image preprocessing and semantic segmentation concepts to satellite imagery for road-network extraction.
- Prepared datasets and evaluated model outputs for infrastructure-mapping use cases.

**Cybersecurity and LLM Security Research** **[GitHub]**
*OWASP | LLM Security | Vulnerability Assessment | AI Security*

- Studied security risks affecting LLM applications, including prompt-injection and data-poisoning scenarios.
- Translated security concepts into practical assessment and remediation workflows for AI applications.

## LANGUAGES

English | Hindi | Bengali